

The Full K-Bound Manuscript, Plainly

A reader’s companion to the 60-page manuscript
“K-Bound: When Is Label-Free Adaptation Knowable?”

Companion narrative to the manuscript of Pratik Niroula

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This is the plain-language companion to the *long* manuscript (`kbound.pdf`, 60 pages). The long version is the project’s complete scientific record — every theorem, every extension, every audit — while the 23-page short paper is the claim-controlled submission core. If you read only one companion, read the short one first; this one covers everything the short paper deliberately left on the shelf. Nothing here changes a claim or hides a caveat.

What the 60-page version is, and is not

The project’s own documentation is direct about the division of labor: the short paper is the submission; the long manuscript is the *source inventory* — the place where the full theory stack, the complete experimental battery, the failed and superseded attempts, and the historical corrections all live under one roof. Think of it as the laboratory notebook promoted to manuscript form: broader, deeper, and more candid than any conference format would allow.

That framing matters for reading it. The long manuscript keeps results the short paper demoted, keeps diagnostics the short paper only summarizes, and keeps a written trail of what was once believed and later corrected. It is longer because the story it tells includes its own revisions.

1 The same spine, told once more in six sentences

[[manuscript: abstract, §1](#)]

A deployed model can self-update on unlabeled data (test-time adaptation), and the same update that heals under one shift silently harms under another — with no labels to tell which. K-Bound asks the pre-question: *should* the model adapt? It answers with three actions: adapt, freeze, or abstain. The theory proves the benefit’s direction is knowable over a declared drift class *iff* an observable margin beats the declared drift budget, $|M| > \beta$; inside the band, evidence-identical worlds with opposite truths make abstention mandatory. The method, KGA, replaces the unobservable population test with a calibrated interval $\hat{\Delta} \pm \varepsilon$ and commits only

when the interval clears zero, capping the false-adapt rate. The evidence shows uniform no-harm on natural shifts, real routing wins where harm is mixed and detectable, and honest nulls everywhere else.

That is the trunk. Everything below is a branch the long manuscript grows from it.

2 The four pillars, and how the long version deepens them

[manuscript: §4–5]

The long manuscript builds the theory as four pillars. **Pillar one:** the disagreement-region reduction — the contest between old and new model lives entirely on the inputs where they disagree, where the benefit’s sign becomes $\text{sign}(M + \gamma)$. **Pillar two:** the impossibility — twin worlds, identical in every label-free observable, opposite in truth; the formal engine is a classical two-point (Le Cam) argument, and the manuscript supplies an explicit numerical witness pair rather than an existence claim. **Pillar three:** the exact frontier $|M| > \beta$ with the three-way rule as the *unique maximal sound* certificate — “maximal” meaning it commits everywhere commitment is defensible and nowhere else. **Pillar four:** the finite-sample certificate, whose false-adapt cap survives with an honest “marginal, not conditional” label.

Two upgrades appear only here. First, **split-observability:** the proofs never actually use anything specific to test-time adaptation — only the structure “sign of the quantity = sign of (observable + bounded hidden term).” Any decision problem with that shape inherits the whole framework; TTA is the demonstrated instance. Second, **unification:** the popular label-free accuracy estimators — ATC, DoC, GDE, COT, AETTA, agreement-on-the-line — are shown to be the $\beta = 0$ face of the frontier. In plain terms: each of those methods is exactly what K-Bound becomes if you assume calibration never drifts. They are not wrong; they are the zero-budget special case, sound precisely when that assumption holds.

3 The extended theory shelf, one plain paragraph each

[manuscript: §“Theory extensions” and companion-stack appendices]

The one-bit dichotomy. Suppose you refuse to declare any budget and ask instead: what is the *least* extra information that makes the benefit sign decidable in the unknowable band? The manuscript proves a dichotomy: no assumption that can be *checked against the evidence* ever suffices — because the twin worlds share all evidence, any checkable assumption holds in both or neither — and the minimal untestable supplement is exactly *one bit*: an orientation, “which way does drift lean.” Nothing smaller works; nothing checkable substitutes. This settles the project’s old Conjecture 1 *negatively*: universal label-free benefit bracketing does not exist.

The weakest one-bit class. Given that one bit must be assumed, on *which* class of worlds is one bit enough? A natural class (margin-monotone calibration) works unconditionally. Is

it *the* weakest such assumption? Answer, and it is the most mathematically distinctive result on the shelf: there is *no unique weakest class*. The unconditionally weakest classes form an explicit *finite family of dominance polytopes* — finitely many tied winners, characterized exactly — with the earlier “general position” condition reappearing as the degenerate face where the family collapses to one member. A machine validator sweeps hundreds of thousands of cases with zero mismatches.

Minimax optimality and exact constants. On the knowable side, could a cleverer rule need fewer samples? No: the certificate’s sample complexity is minimax order-optimal, and in a canonical three-world configuration the manuscript pins the *exact constant* $\kappa(\alpha)$ — not just the rate but the number.

Anytime validity. Real deployments do not score one batch; they watch a stream and stop whenever they like. Naively re-testing forever destroys error control. Using a betting-supermartingale (Ville-type) construction, the streaming certificate holds its false-adapt cap *time-uniformly under optional stopping*: valid whenever you stop, even adversarially.

Multicandidate routing. With several adapters proposing candidates at once, an adversarial selector could shop for the one that slipped through. A Bonferroni-corrected construction makes the guarantee *family-wise*: the cap holds over the whole panel, no matter how the winner is chosen. Multiclass versions of both extensions are proved as well.

Multiclass capacity impossibility. Hope: one scalar “capacity” summarizing multiclass adaptability. Proof: no such single scalar exists in the general vector-concept regime. Another door closed honestly.

Margin computability and regression. A dichotomy sorts out exactly when the frontier margin is computable from stated structure, and the regression analogue proves a bounded-drift bracketing iff-condition together with the matching general impossibility.

Auditable budgets. The budget β must be declared — but can it at least be *audited*? Provably not for free: label-free audits are vacuous over the unrestricted class. With purchasable structure — a small labeled audit set, a Lipschitz transport assumption, or an index-drift model — a computed upper bound $\hat{\beta}$ with finite-sample validity becomes available, and the manuscript states exactly when it may replace the declared budget. Assumptions in, guarantees out; nothing for nothing.

4 The experimental record, wider and with its corrections shown

[manuscript: §6–7, “Excluded wins,” supplementary panels]

The long manuscript runs the same regime logic as the short paper across a wider battery — the full CIFAR-10-C corruption suite alongside the stress grid, online non-stationary streams where always-adapt visibly collapses, ten-backbone ImageNet-R sweeps, the complete PACS and

RxRx1 tables, the constructed cross-protocol aggregate — and then does something conference papers almost never do: it prints its own case files.

A section titled *Excluded wins and regime boundaries* lists, one by one, the results that once looked like victories and why each was demoted. The early Office-Home and iWildCam “beats-both” wins: traced to an in-sample calibration radius — a bug — corrected to no-harm under the proper out-of-fold radius. The Camelyon17 pooled win: traced to in-distribution cells contaminating the held-out pool; on genuine out-of-distribution data it evaporates, and the manuscript says so. A CIFAR-10-C SAR aggregate that would not reproduce from raw files: quarantined outright, contributing nothing. Multicandidate routes that flag wins at a 78% pooled false-adapt rate: displayed as the cautionary tale they are. The manuscript even preserves the earlier, weaker single-seed ImageNet-C configuration alongside its five-seed successor for provenance. (One bookkeeping consequence of that layered history: the long manuscript’s front table still carries the superseded single-seed ImageNet-C numbers with a footnote, while the short paper and the generated manifest carry the current five-seed values — worth syncing on the next editing pass.)

Read as a whole, the experimental chapters make a quiet argument: the headline numbers are believable *because* the graveyard is public.

5 The machine-checked layer

[manuscript: reproducibility and Lean appendices]

Core steps of the theory are formalized in Lean 4 — the frontier’s deterministic branches, algebraic witnesses for the bands, the coverage-to-false-adapt implication, and, in later waves, paper-faithful foundations (an exchangeable-score conformal reduction, a discrete Ville step, two-point packaging). The manuscript is precise in both directions: no **sorrys** and no project axioms in what is checked, and an explicit list of what is *not* formalized (the measure-theoretic witness constructions, risk alignment, calibration transfer, and more), with no claim that a full probability library was rebuilt and no priority claim. The formal artifact is offered as scaffolding for trust, not as a trophy.

6 How to read the 60 pages

A practical route through the manuscript: read the abstract and §1 for the full-throated version of the story (the long abstract is more explicit than the short paper’s about what is inherited from prior work versus sharpened here); skim §2’s related-work positioning; take §4–5 as the four pillars with proofs; treat the experiments the way this companion did — synthetic validation first, stress grids, then natural shifts, then the *Excluded wins* section, which is the manuscript’s conscience; and dip into the extension appendices as independent short stories, each with its own stated assumptions. The limitations section’s numbered points (1)–(8) double

as the project’s own honest self-review, including the standing admission that no independent external replication is claimed yet.

If you want...	Go to
the decision problem and frontier, fully proved	§4–5 (four pillars)
why abstention is mandatory, with a witness	impossibility pillar + witness figures
the “no free lunch” for assumptions	one-bit dichotomy appendix
the exactly-characterized weakest assumptions	weakest-class appendix (dominance polytopes)
streaming and many-adapter guarantees	anytime + multicandidate extensions
optimality (nobody can beat this rate)	minimax + exact-constants appendix
the full evidence panel with tiers	experiments chapters + uniform panel
the demoted and withdrawn results, explained	<i>Excluded wins and regime boundaries</i>
what a proof assistant actually verified	Lean formalization appendix

Table 1: *
A map of the long manuscript by reader intent.

The one-paragraph verdict

The long manuscript is the complete K-Bound: the exact boundary between knowable and unknowable adaptation, the price list for crossing it (one untestable bit, or a declared and auditable budget), the certificate that operationalizes it with a capped error rate, optimality results saying no method of this kind can do fundamentally better, streaming and multi-adapter generalizations, a wide empirical panel graded by evidence tier — and, running through all of it, a documented habit of demoting its own results when the evidence said so. It is not a submission; it is the record a submission stands on.